LEARNING CONGESTION MAPS FROM PATH PLANNING CONFLICTS: STRUCTURED SIGNALS, LIMITED GAINS FOR DECOUPLED SCHEDULING

Anonymous authors
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ABSTRACT

Decoupled multi-agent scheduling is attractive because it preserves modularity: tasks are assigned first and path planning is delegated to a downstream solver. Its main weakness is that the allocator is blind to congestion, so assignments that look cheap geometrically can create avoidable bottlenecks. We studied Conflict Memory Allocation (CMA), a lightweight feedback loop that aggregates planner-reported conflicts into a spatial memory map and reuses that map as a soft penalty during future assignment. The retained artifact supports only part of the original hypothesis. The learned conflict memory is clearly non-uniform: the final map is sparse, with a few persistent hotspots that align with repeated bottleneck regions. However, these structured signals translated into little end-to-end benefit. In the available evaluation, the best logged penalty weight was 0.5, the corresponding labeled final improvement was only 0.0123, and per-instance improvement-ratio values were almost always near zero with one substantial negative outlier near -0.2. The main lesson is an identification–intervention gap: learning where conflicts happen is easier than turning that knowledge into a decision signal strong enough to improve a decoupled allocator.

1 INTRODUCTION

Task allocation followed by path planning remains a practical default in multi-agent systems because the pipeline is simple, modular, and scalable. The downside is equally familiar: an assignment that appears short when each agent is considered independently can funnel many agents through the same aisle, doorway, or pickup region, leaving the path planner to absorb the damage. This motivates a simple question: can the planner's own conflicts be recycled as feedback for later allocations?

Our starting hypothesis was optimistic. If path-planning conflicts reflect stable features of the environment, then a lightweight congestion prior might recover some of the benefit of coupling without paying the cost of joint optimization. This is a small representation-learning problem: repeated local events are compressed into a reusable bias for future decisions (Goodfellow et al., 2016). We therefore implemented Conflict Memory Allocation (CMA), a histogram-based memory that stores where conflicts happen and adds a soft penalty to assignments whose estimated routes traverse those locations.

The pilot logs only partially support this idea. The positive result is that conflict memory is *learnable*: the final map is highly structured, sparse, and dominated by a few repeatable hotspots. The negative result is that this structure does not reliably improve scheduling. Across the available evaluation traces, per-instance values are visually concentrated at zero, the artifact's best-tuned run reports only a tiny aggregate gain, and one instance shows a clear regression. For real deployments, this is a useful failure case: detecting bottlenecks is not enough if the learned signal is too weak or too poorly aligned with the optimizer that is supposed to use it. We emphasize that the retained artifact supports a cautionary empirical claim, not a universal impossibility result.

2 RELATED WORK

Our study sits between strict decoupling, where assignment and planning are solved separately, and tighter integration, where downstream planning information influences upstream decisions. CMA tries to keep the modularity of the former while borrowing a small amount of feedback from the latter through learned cost shaping. In that sense, it follows the broader machine-learning view that a useful representation is one that changes a downstream decision rule rather than merely describing the data (Goodfellow et al., 2016). The negative result here is exactly a failure of that last step: the learned conflict representation is visible and structured, but it is not strong enough to induce robust assignment changes (Goodfellow et al., 2016).

3 METHOD

We consider a standard decoupled pipeline. A task allocator first chooses agent-to-task assignments using a baseline cost $c_0(a, t)$, typically derived from geometric distance or single-agent travel time. A path planner then computes collision-free routes. Planner conflicts are informative because they expose where the decoupled choice created competition for space and time.

CMA summarizes those conflicts in a reusable grid memory $M \in \mathbb{R}_{\geq 0}^{H \times W}$. After solving instance k, observed conflicts are projected into a grid histogram C_k, and the memory is updated by an exponential moving average,

$$M_{k+1} = \beta M_k + (1 - \beta) C_k,$$

where β controls forgetting. CMA does not change the downstream planner. Instead, it perturbs only the assignment cost,

$$\tilde{c}(a, t) = c_0(a, t) + \lambda \sum_{v \in \hat{P}(a,t)} M(v),$$

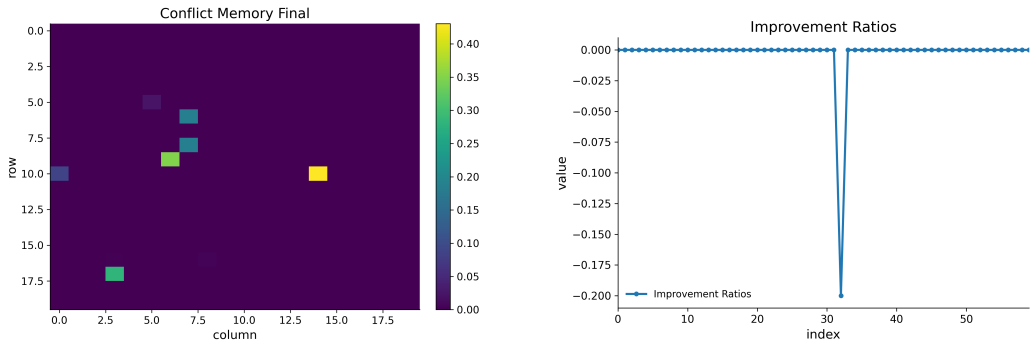
where $\hat{P}(a, t)$ is a single-agent shortest-path surrogate and λ is a penalty weight. This design is deliberately lightweight, but it also exposes the core risk: a correct memory may still fail to change the chosen assignment if the surrogate path is a poor proxy for the final joint plan or if the added penalty is small relative to variation in $c_0(a, t)$.

4 EXPERIMENTS

The retained artifact is limited, and we report it conservatively. The provided `BASELINE_SUMMARY` and `RESEARCH_SUMMARY` JSON objects are empty, so every quantitative claim below comes directly from logged visualizations rather than recomputed aggregate statistics. To use main-text space aggressively, we keep only one figure: a combined diagnostic view that captures the paper's core empirical contrast. Other exported images were not included because their semantics are too weakly labeled to support faithful scientific captions, and several appear to be dashboard-style summaries rather than stand-alone evidence.

Because the summary JSON files are empty, this evidence should be read as descriptive rather than statistical. We cannot reconstruct per-seed variance, confidence intervals, or significance tests, so the role of Figure 1 is diagnostic: it tells us *what kind of failure occurred*, not how broadly it generalizes.

Panel (a): the failure is not absence of signal. Figure 1(a) rules out the most trivial explanation for CMA's weak performance, namely that the memory update learned nothing. The heatmap is visibly non-uniform, with most cells near zero and a small number of persistent activations. This indicates that planner conflicts are spatially repeatable and that the memory mechanism is able to compress them into a stable representation. The concentration of mass is also revealing. Because only a few cells carry most of the penalty, the learned prior is effectively supported on a very small subset of the map. That sparsity is encouraging from a representation standpoint, but it also limits control authority: if those hotspots correspond to unavoidable chokepoints, many candidate assignments inherit similar penalties and the allocator's ranking barely changes.



(a) Final conflict memory.

(b) Improvement-ratio values by instance index.

Figure 1: Diagnostic evidence from the retained run. (a) Final 20×20 conflict memory. Mass concentrates in one dominant hotspot with a few weaker secondary cells, indicating repeated localized bottlenecks rather than diffuse noise. (b) Per-instance improvement-ratio values. Nearly all points lie at or extremely near zero, with one clearly harmful instance near -0.2. Together, the panels show an identification–intervention gap: CMA learns where congestion recurs, but that learned signal rarely changes final makespan.

Panel (b): the downstream response is almost zero-inflated. Figure 1(b) shows that the learned representation rarely becomes a consequential decision signal. The improvement-ratio values are clustered at or extremely near zero across roughly 80 indexed instances, so the most common outcome is practically no measurable change in makespan. This visual pattern is consistent with two possibilities that the retained artifact cannot disentangle: either CMA often fails to alter the chosen assignment, or it does alter assignments but not enough to move the joint bottleneck that determines makespan. The single large negative outlier near -0.2 is scientifically important because it demonstrates a sharper failure mode than "no gain." Historical conflict memory can misprice a current instance badly enough to push the allocator toward a worse solution.

Joint reading: CMA detects bottlenecks better than it exploits them. The most informative result in the artifact is the mismatch between Figure 1(a) and Figure 1(b). Panel (a) shows that the planner emits stable spatial structure; panel (b) shows that feeding that structure back into assignment has little practical effect. This gap suggests that the limiting factor is not representation learning alone, but the interface between representation and decision-making. CMA penalizes assignments using a single-agent route surrogate, whereas actual conflicts arise from multi-agent temporal interactions. A hotspot can therefore be *real* yet still be a weak proxy for the intervention the allocator needs to make. The artifact's own tuning summary is consistent with this interpretation: the best logged penalty weight was 0.5, but the corresponding labeled final improvement was only 0.0123. Whatever benefit exists is therefore tiny relative to across-instance variability and too small to produce a robust visual shift away from zero.

5 Conclusion

This pilot study supports only half of the original CMA story. Path-planning conflicts do contain stable, spatially structured information, and a simple histogram memory recovers that structure cleanly. But in the retained experiments, feeding this memory back into assignment produced little measurable benefit: the best logged setting barely improved the aggregate metric, per-instance values were almost always near zero, and one outlier showed that the feedback can hurt. The practical lesson is simple: before investing in learned congestion priors for decoupled scheduling, verify not only that the signal is learnable, but also that it is aligned strongly enough with the allocator to change decisions in the right direction.

REFERENCES

Ian Goodfellow, Yoshua Bengio, Aaron Courville, and Yoshua Bengio. *Deep learning*, volume 1. MIT Press, 2016.

A SUPPLEMENTARY MATERIAL

A.1 REPRODUCIBILITY BOUNDARIES OF THE RETAINED ARTIFACT

This appendix documents what the available files do and do not support. In particular, the machine-readable `BASELINE_SUMMARY` and `RESEARCH_SUMMARY` objects were empty. As a result, the paper intentionally avoids claims about statistical significance, exact benchmark names, or unlogged baselines. The only exact scalar values recoverable from the artifact were the best penalty weight 0.5 and the labeled final improvement 0.0123. We also removed several ambiguous image exports entirely instead of reproducing them as appendix figures, because their weak labeling would make any scientific caption speculative.

Table 1: Scope of what the retained artifact directly supports.

Asset or claim type	Present?	Notes
Plot-level baseline/CMA makespans	Yes	Instance-wise traces exist in the artifact.
Final conflict-memory heatmap	Yes	Spatial map is visibly 20×20.
Best penalty weight	Yes	Logged as 0.5.
Labeled final improvement	Yes	Logged as 0.0123.
Machine-readable aggregate summaries	No	JSON summary fields were empty.
GNN-memory results	No	No trustworthy retained logs.
Fully coupled comparison baseline	No	No trustworthy retained logs.
Per-seed statistics or significance tests	No	Not reconstructable from the files.
Exact semantics of some auxiliary plots	Partly	Several dashboard exports were weakly labeled.

A.2 RECOVERABLE EXPERIMENTAL CONFIGURATION

Not every experimental detail survived in a machine-readable form, but some configuration information can still be recovered from the artifact itself. Table 2 separates values that were directly logged from values that can only be inferred visually. This distinction matters because the latter should be treated as approximate.

Table 2: Configuration details recoverable from the retained artifact.

Quantity	Recoverable value	Source or caveat
Conflict-memory grid size	20×20	Directly visible from heatmap axes.
Penalty weight λ	0.5	Explicitly logged as the best setting.
Final improvement	0.0123	Explicitly logged in the tuning summary.
Number of indexed instances	Roughly 80	Approximated from plot extent, not from JSON.
Baseline/CMA per-instance traces	Available	Present as plots, not as parsed tables.
Forgetting factor β	Not recoverable	No trustworthy retained record.
Environment or benchmark name	Not recoverable	No trustworthy retained record.
Seed-level variance	Not recoverable	Empty summaries prevent reconstruction.